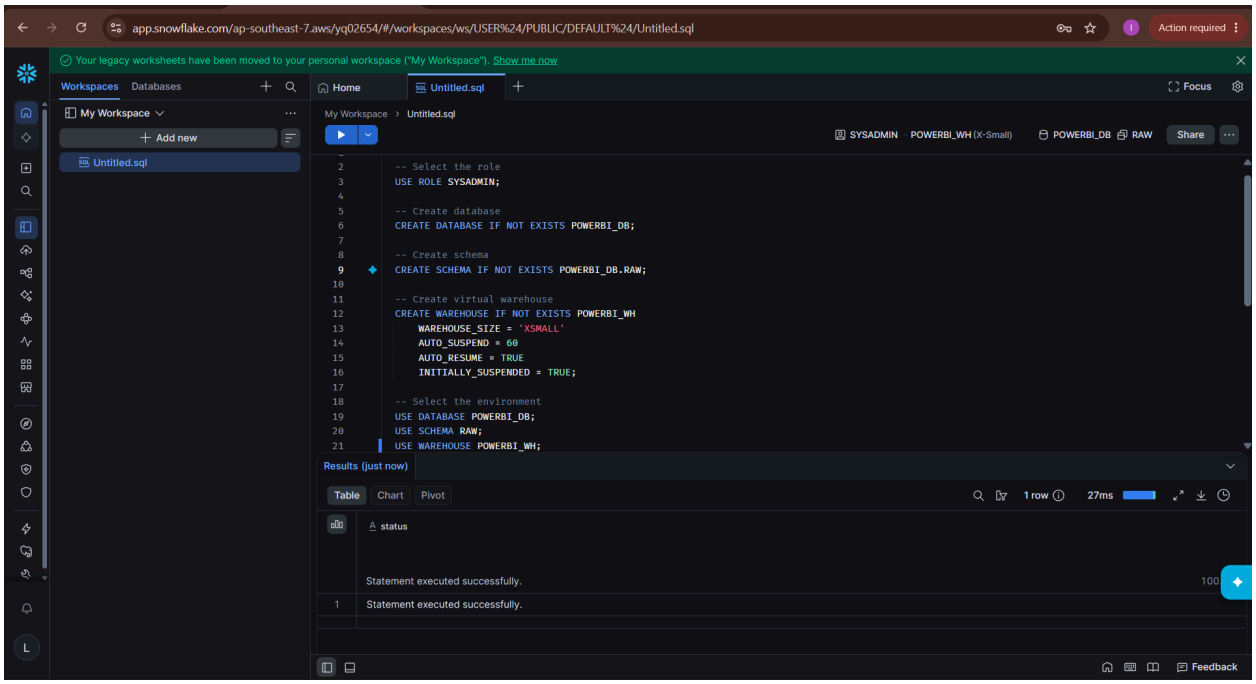
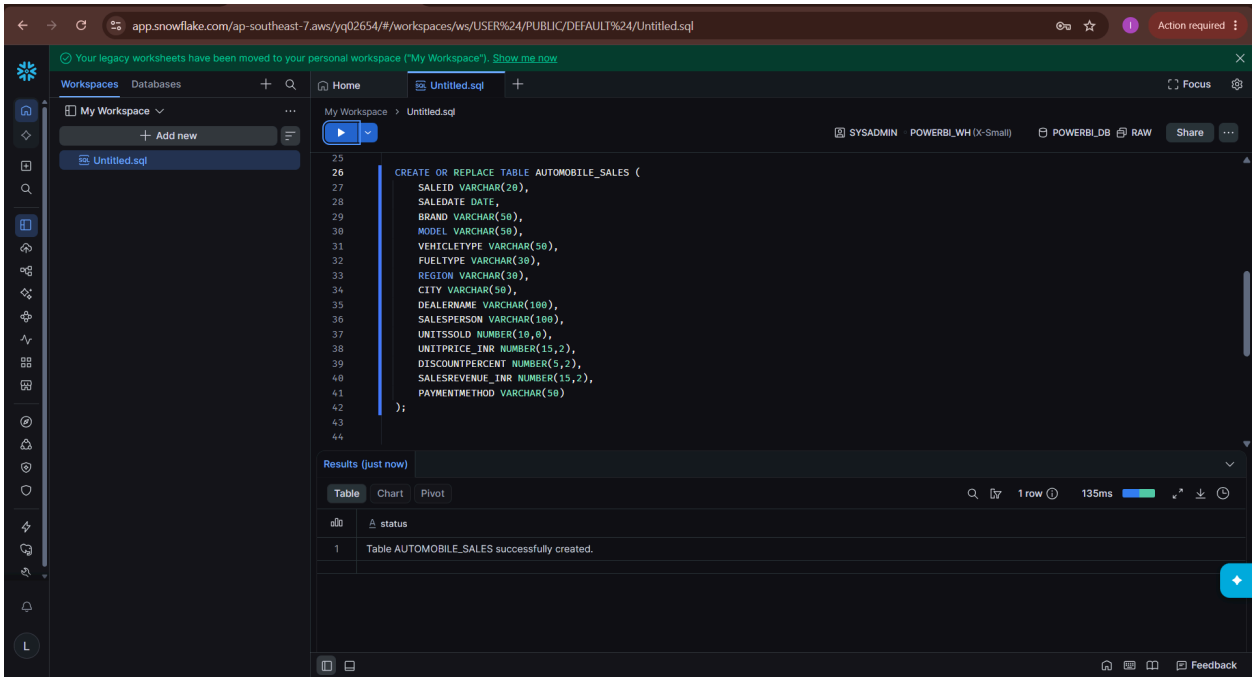
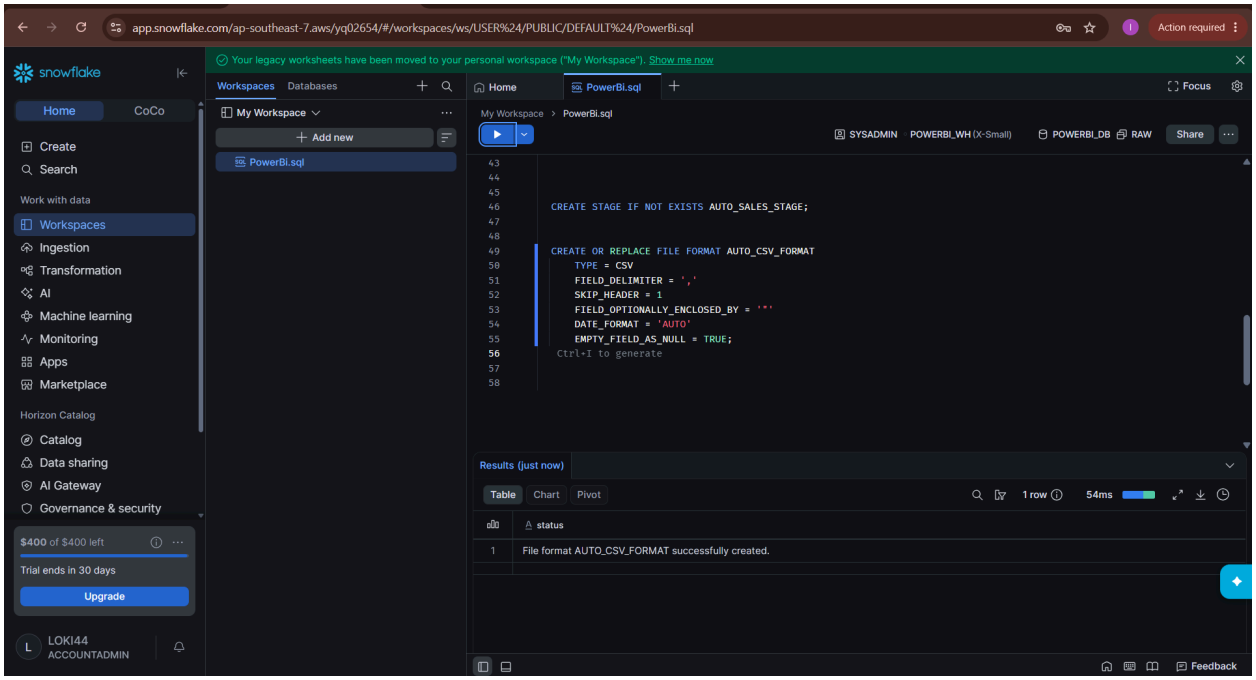
Create the Snowflake environment

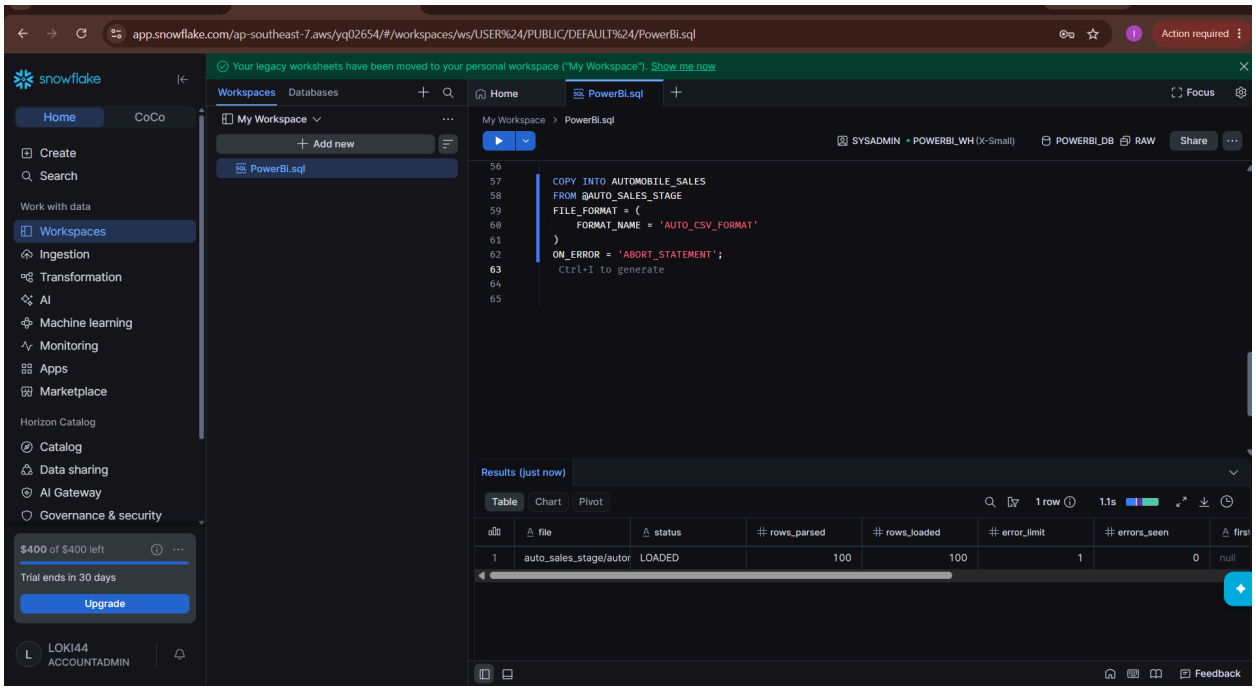


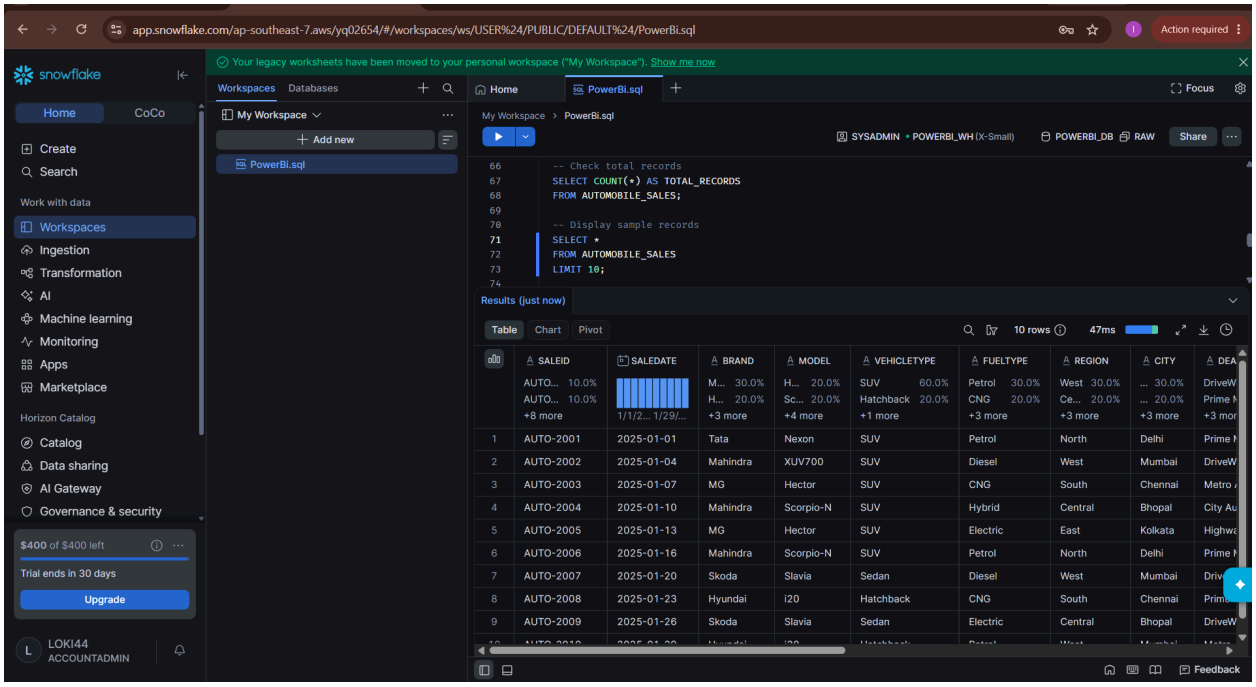
Create the automobile sales table



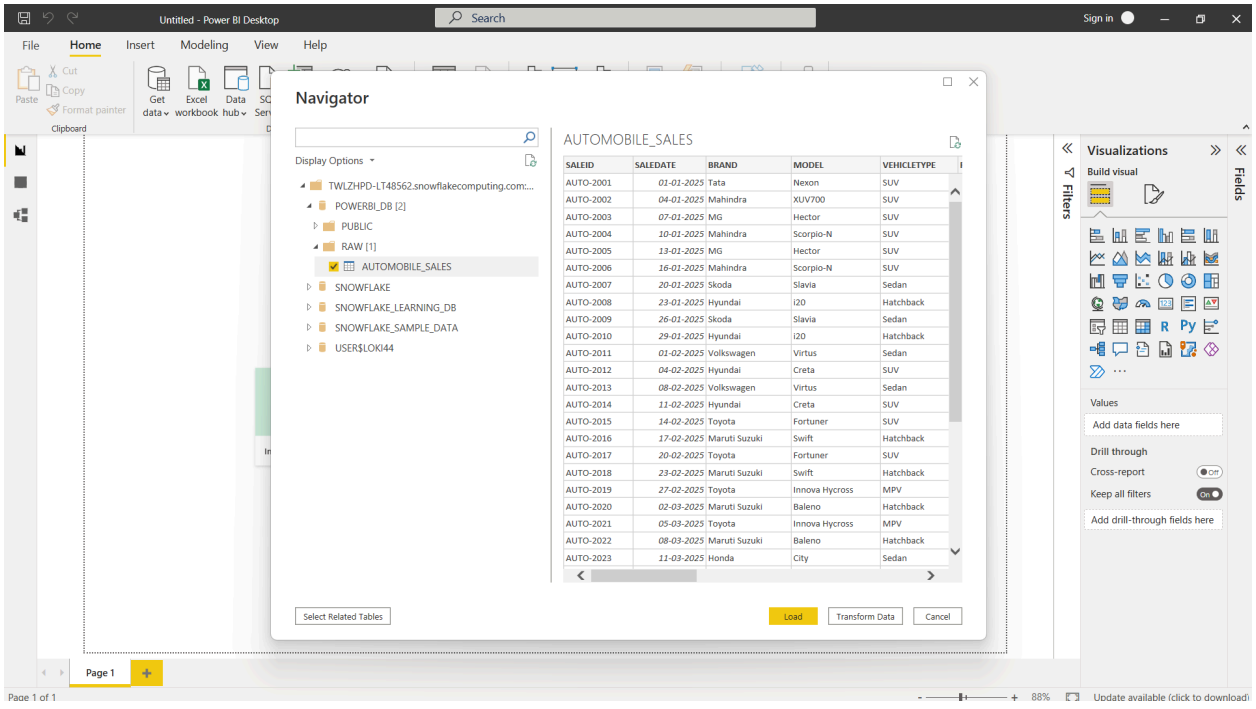
Validate the data in Snowflake


app.snowflake.com/ap-southeast-7.aws/yq02654/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/PowerBi.sql
Action required
snowflake
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
Home
PowerBi.sql
Focus
Home
CoCo
My Workspace
Add new
PowerBi.sql
Create
Search
Work with data
Workspaces
Ingestion
Transformation
AI
Machine learning
Monitoring
Apps
Marketplace
Horizon Catalog
Catalog
Data sharing
AI Gateway
Governance & security
My Workspace › PowerBi.sql
SYSADMIN · POWERBI_WH (X-Small)
POWERBI_DB
RAW
Share
43
44
45
46 CREATE STAGE IF NOT EXISTS AUTO_SALES_STAGE;
47
48
49 CREATE OR REPLACE FILE FORMAT AUTO_CSV_FORMAT
50 TYPE = CSV
51 FIELD_DELIMITER = ','
52 SKIP_HEADER = 1
53 FIELD_OPTIONALLY_ENCLOSED_BY = '"'
54 DATE_FORMAT = 'AUTO'
55 EMPTY_FIELD_AS_NULL = TRUE;
56 Ctrl+I to generate
57
58
Results (just now)
Table
Chart
Pivot
1 row
54ms
status
1
File format AUTO_CSV_FORMAT successfully created.
$400 of $400 left
Trial ends in 30 days
Upgrade
LOKI44
ACCOUNTADMIN
Feedback

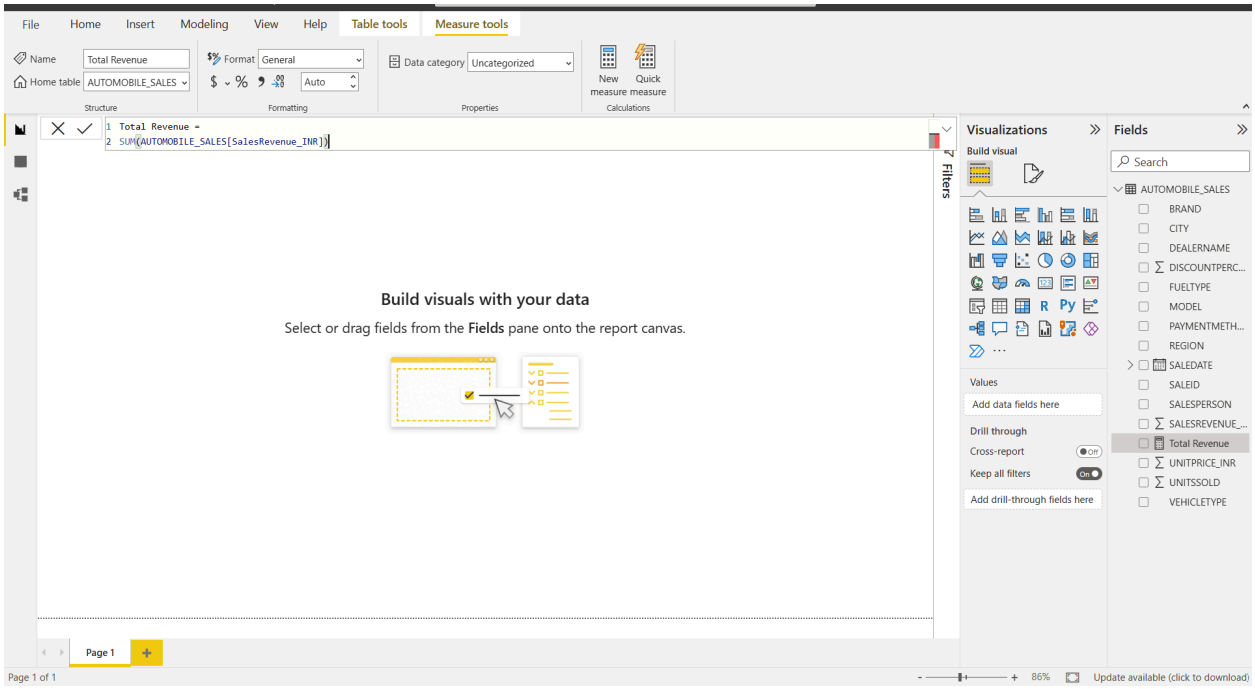

app.snowflake.com/ap-southeast-7.aws/yq02654/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/PowerBi.sql
Action required
snowflake
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
Home
PowerBi.sql
Focus
Home
CoCo
My Workspace
Add new
PowerBi.sql
Create
Search
Work with data
Workspaces
Ingestion
Transformation
AI
Machine learning
Monitoring
Apps
Marketplace
Horizon Catalog
Catalog
Data sharing
AI Gateway
Governance & security
My Workspace › PowerBi.sql
SYSADMIN · POWERBI_WH (X-Small)
POWERBI_DB
RAW
Share
56
57 COPY INTO AUTOMOBILE_SALES
58 FROM @AUTO_SALES_STAGE
59 FILE_FORMAT = (
60 FORMAT_NAME = 'AUTO_CSV_FORMAT'
61)
62 ON_ERROR = 'ABORT_STATEMENT';
63 Ctrl+I to generate
64
65
Results (just now)
Table
Chart
Pivot
1 row
1.1s
file
status
rows_parsed
rows_loaded
error_limit
errors_seen
1
auto_sales_stage/autor
LOADED
100
100
1
0
null
$400 of $400 left
Trial ends in 30 days
Upgrade
LOKI44
ACCOUNTADMIN
Feedback

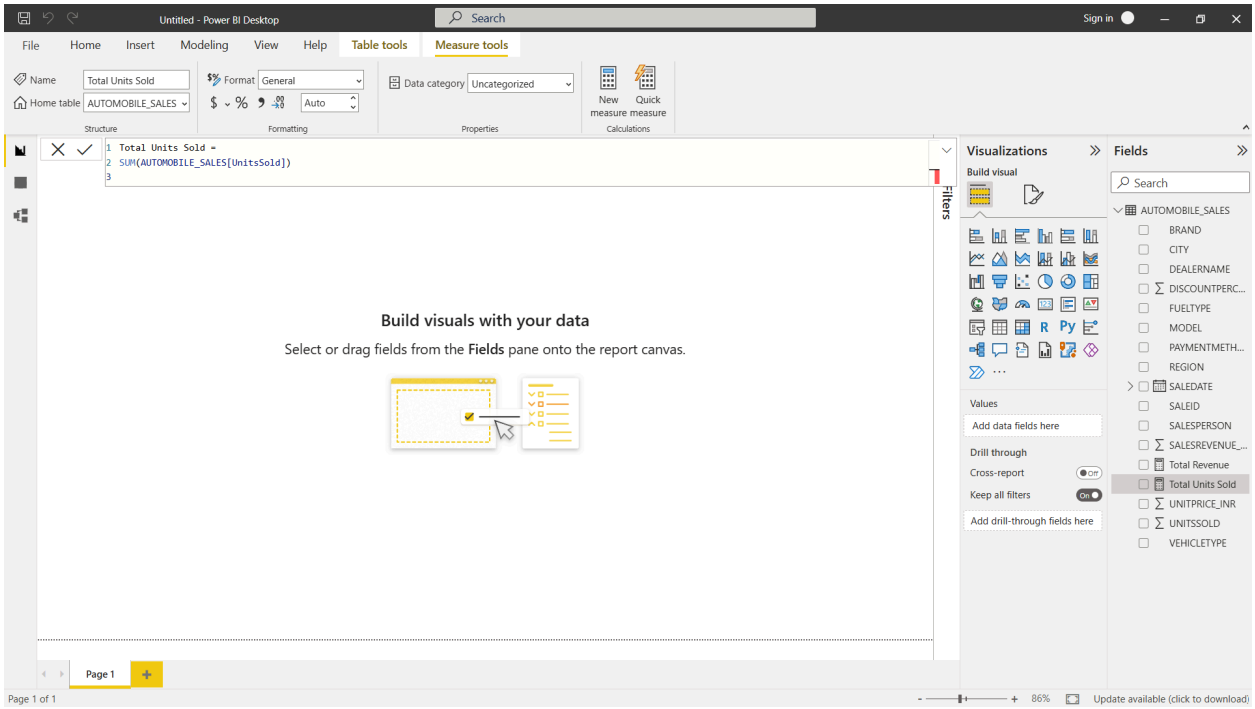


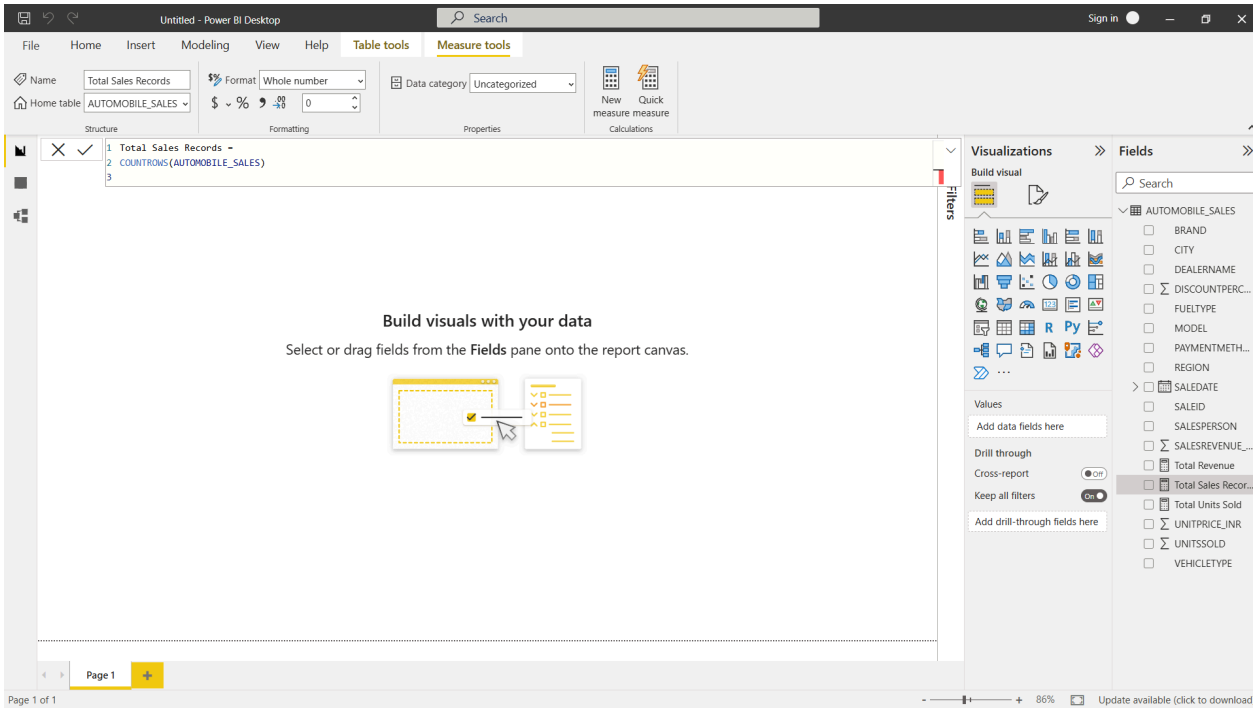
Connect Snowflake to Power BI and Load the Data

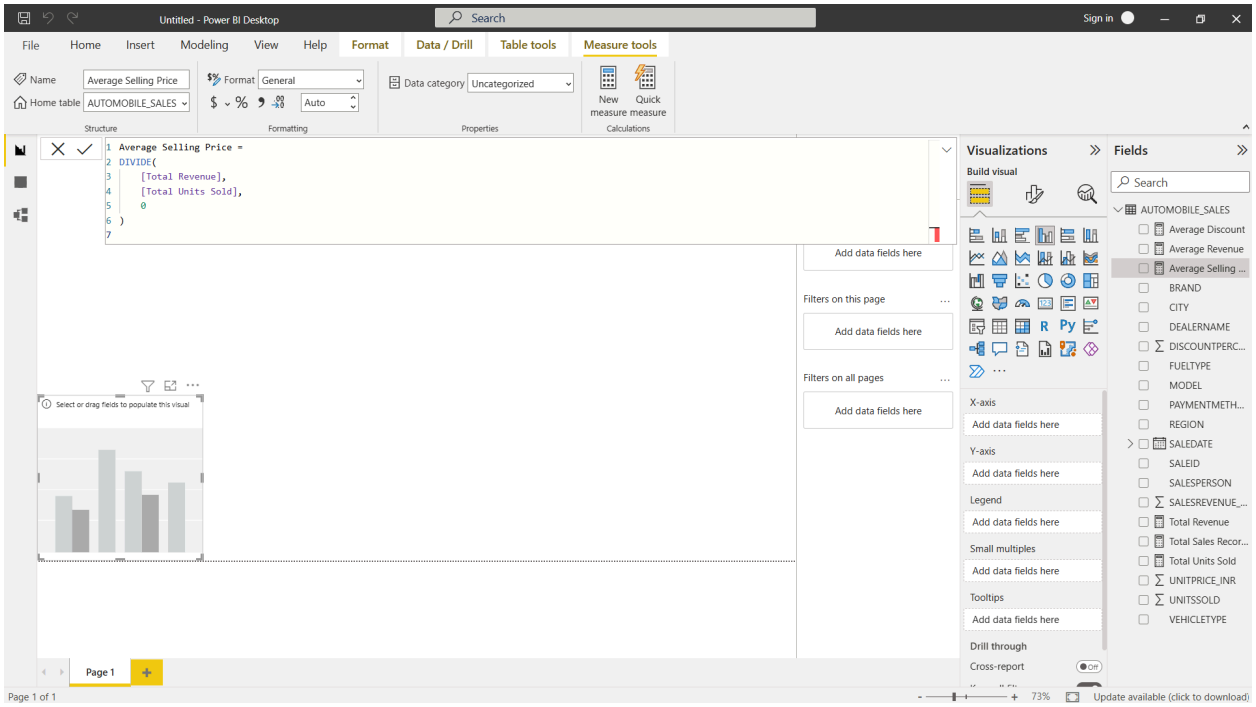


Create DAX measures


File
Home
Insert
Modeling
View
Help
Table tools
Measure tools
Name
Total Revenue
Home table
AUTOMOBILE_SALES
Structure
Format
General
Auto
Formatting
Data category
Uncategorized
Properties
New measure
Quick measure
Calculations
1 Total Revenue =
2 SUM(AUTOMOBILE_SALES[SalesRevenue_INR])
Build visuals with your data
Select or drag fields from the Fields pane onto the report canvas.
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Fields
Search
AUTOMOBILE_SALES
BRAND
CITY
DEALERNAME
DISCOUNTPERC...
FUELTYPE
MODEL
PAYMENTMETH...
REGION
SALEDATE
SALEID
SALESPERSON
SALESREVENUE_...
Total Revenue
UNITPRICE_INR
UNITSSOLD
VEHICLETYPE
Filters
Page 1
Page 1 of 1
86%
Update available (click to download)


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
View
Help
Table tools
Measure tools
Name
Total Units Sold
Home table
AUTOMOBILE_SALES
Structure
Format
General
Auto
Formatting
Data category
Uncategorized
Properties
New measure
Quick measure
Calculations
1 Total Units Sold =
2 SUM(AUTOMOBILE_SALES[UnitsSold])
3
Build visuals with your data
Select or drag fields from the Fields pane onto the report canvas.
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Fields
Search
AUTOMOBILE_SALES
BRAND
CITY
DEALERNAME
DISCOUNTPERC...
FUELTYPE
MODEL
PAYMENTMETH...
REGION
SALEDATE
SALEID
SALESPERSON
SALESREVENUE_...
Total Revenue
Total Units Sold
UNITPRICE_INR
UNITSSOLD
VEHICLETYPE
Filters
Page 1
Page 1 of 1
86%
Update available (click to download)





7. Dashboard Development

The dashboard was developed using Microsoft Power BI Desktop.

The report includes KPI cards, categorical visualizations, trend analysis charts, and interactive filtering options.

7.1 KPI Cards

Four KPI cards were created to display important automobile sales metrics.

Total Revenue: Displays the overall revenue generated from automobile sales.

Total Units Sold: Displays the total number of vehicles sold.

Total Sales Records: Displays the total number of sales transactions available in the dataset.

Average Selling Price: Displays the average revenue per vehicle sold, calculated using total revenue divided by total units sold.

These KPI cards provide a quick overview of overall sales performance.

7.2 Visualizations

The following visualizations were created in Power BI.

1. Line Chart — Monthly Revenue Trend

The line chart displays automobile sales revenue over time. It helps identify fluctuations and patterns in sales performance throughout the dataset period.

2. Card Visual — Total Revenue

The card visual displays the total revenue generated from automobile sales. It provides an immediate summary of the overall business performance.

3. Treemap — Revenue by Brand and Model

The treemap displays revenue contribution using rectangular areas. Each rectangle represents a brand or vehicle model, with the size of the rectangle indicating its revenue contribution.

4. Area Chart — Revenue Trend Over Time

The area chart represents changes in automobile sales revenue over time. The filled area makes it easier to visualize the overall sales trend.

5. Scatter Chart — Vehicle Price vs Units Sold

The scatter chart compares vehicle prices with the number of units sold. It helps identify relationships between vehicle pricing and sales quantity.

6. Waterfall Chart — Brand-wise Revenue Contribution

The waterfall chart displays the contribution of different automobile brands to overall revenue. It helps understand the relative contribution of each brand to total sales.

8. Interactive Slicers and Filters

Interactive slicers were implemented to enable users to explore the dashboard dynamically.

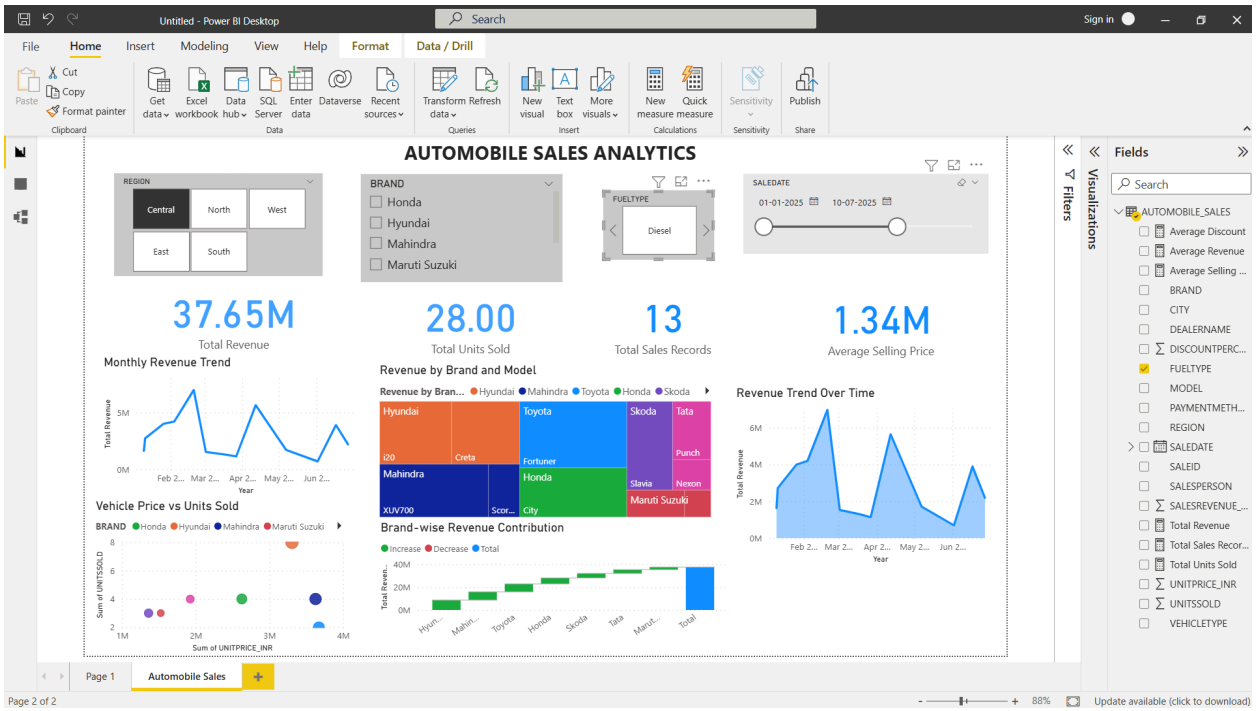
The following slicers were configured:

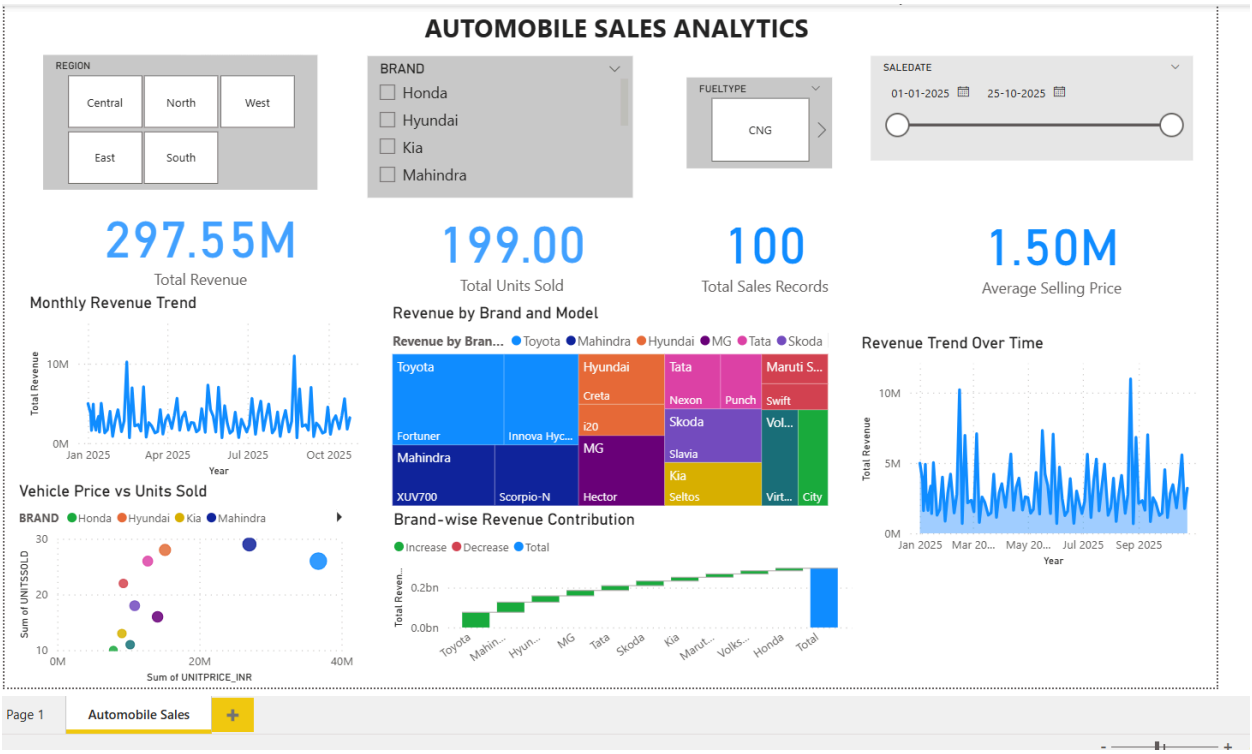
- Region slicer: Filters automobile sales by region.
- Brand slicer: Filters the dashboard based on automobile manufacturers.
- Fuel Type slicer: Filters sales based on fuel categories.
- Date slicer: Filters sales using a selected date range.

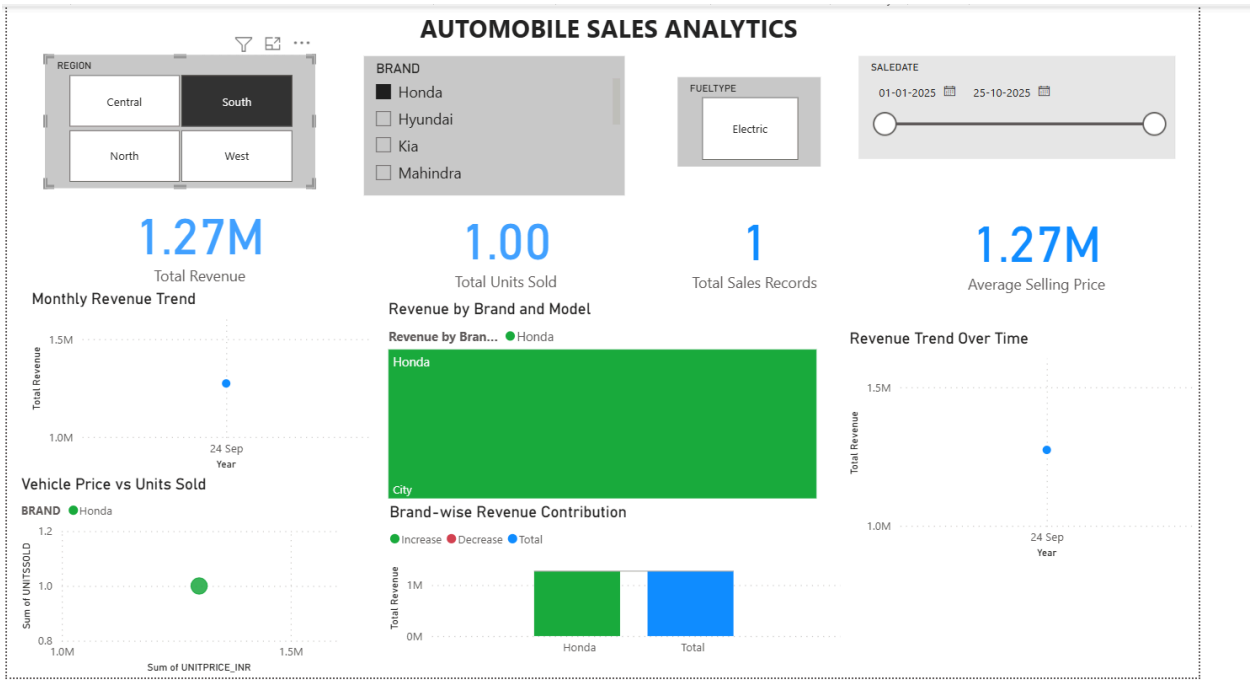
Numeric range slicers can also be used to filter records based on vehicle price or sales revenue.

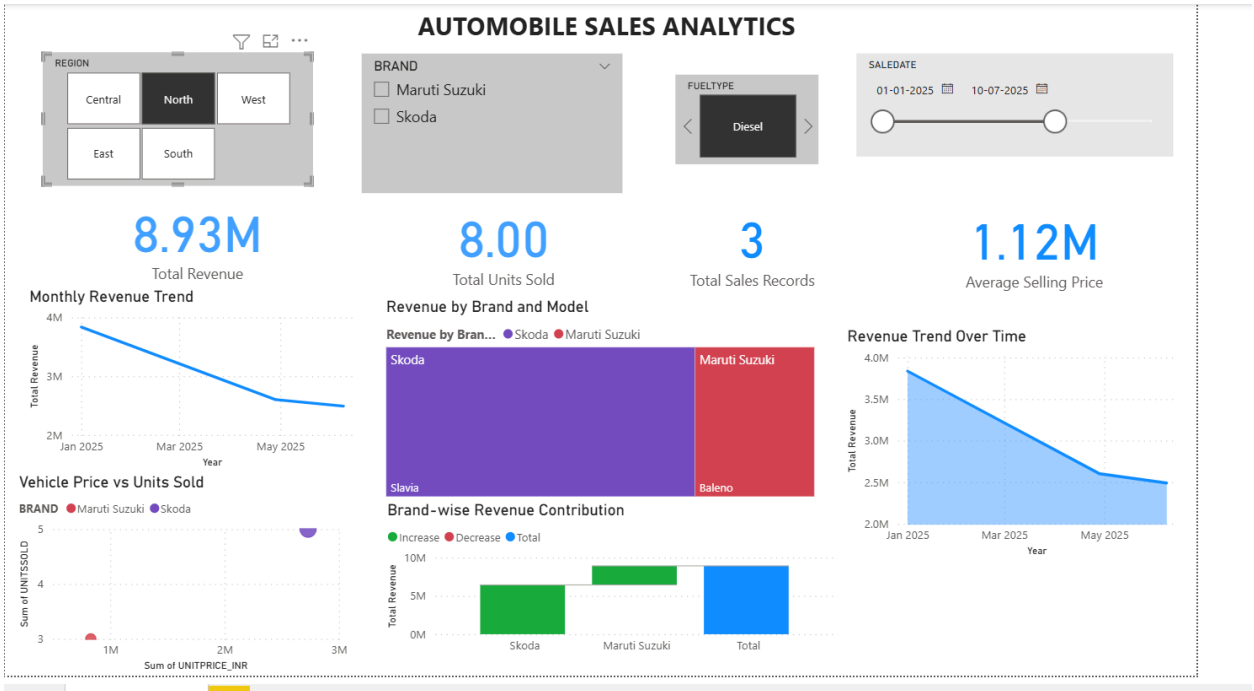
The slicers allow users to dynamically explore the data and analyze sales performance under different conditions.

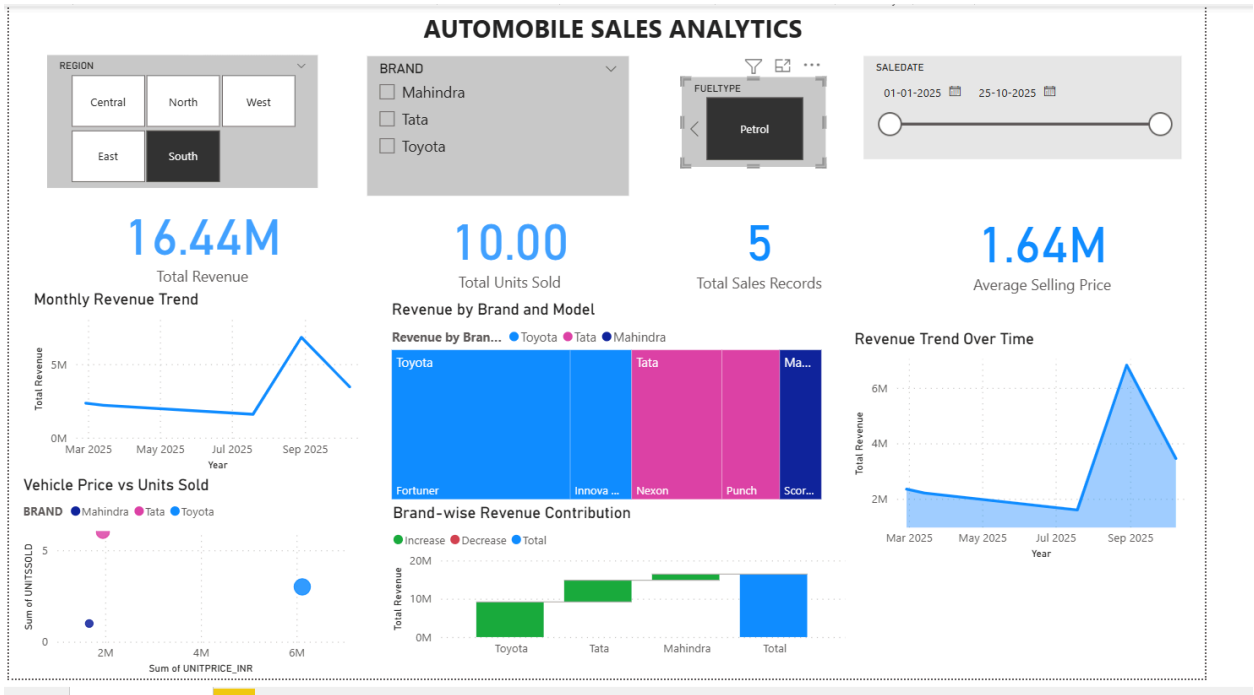
The interactions between slicers and visualizations were configured to ensure that the relevant charts update according to the selected filters.




AUTOMOBILE SALES ANALYTICS
REGION
Central
North
West
East
South
BRAND
Honda
Hyundai
Kia
Mahindra
FUELTYPE
CNG
SALEDATE
01-01-2025
25-10-2025
297.55M
Total Revenue
199.00
Total Units Sold
100
Total Sales Records
1.50M
Average Selling Price
Monthly Revenue Trend
Total Revenue
10M
0M
Jan 2025
Apr 2025
Jul 2025
Oct 2025
Year
Revenue by Brand and Model
Revenue by Bran...
Toyota
Mahindra
Hyundai
MG
Tata
Skoda
Fortuner
Innova Hyc...
Creta
i20
Nexon
Punch
Maruti S...
Swift
Skoda
Slavia
Vol...
Virt...
City
XUV700
Scorpio-N
Hector
Kia
Seltos
Revenue Trend Over Time
10M
5M
0M
Jan 2025
Mar 20...
May 20...
Jul 2025
Sep 2025
Year
Vehicle Price vs Units Sold
BRAND
Honda
Hyundai
Kia
Mahindra
Sum of UNITSSOLD
30
20
10
0M
20M
40M
Sum of UNITPRICE_INR
Brand-wise Revenue Contribution
Increase
Decrease
Total
0.2bn
0.0bn
Toyota
Mahin...
Hyun...
MG
Tata
Skoda
Kia
Marut...
Volks...
Honda
Total
Page 1
Automobile Sales


AUTOMOBILE SALES ANALYTICS
REGION
Central
South
North
West
BRAND
Honda
Hyundai
Kia
Mahindra
FUELTYPE
Electric
SALEDATE
01-01-2025
25-10-2025
1.27M
Total Revenue
1.00
Total Units Sold
1
Total Sales Records
1.27M
Average Selling Price
Monthly Revenue Trend
Total Revenue
1.5M
1.0M
24 Sep
Year
Revenue by Brand and Model
Revenue by Bran...
Honda
Honda
City
Revenue Trend Over Time
1.5M
1.0M
24 Sep
Year
Vehicle Price vs Units Sold
BRAND
Honda
Sum of UNITSSOLD
1.2
1.0
0.8
1.0M
1.5M
Sum of UNITPRICE_INR
Brand-wise Revenue Contribution
Increase
Decrease
Total
1M
0M
Honda
Total


AUTOMOBILE SALES ANALYTICS
REGION
Central
North
West
East
South
BRAND
Maruti Suzuki
Skoda
FUELTYPE
Diesel
SALEDATE
01-01-2025
10-07-2025
8.93M
Total Revenue
8.00
Total Units Sold
3
Total Sales Records
1.12M
Average Selling Price
Monthly Revenue Trend
Total Revenue
4M
3M
2M
Jan 2025
Mar 2025
May 2025
Year
Revenue by Brand and Model
Revenue by Bran...
Skoda
Maruti Suzuki
Skoda
Maruti Suzuki
Slavia
Baleno
Revenue Trend Over Time
Total Revenue
4.0M
3.5M
3.0M
2.5M
2.0M
Jan 2025
Mar 2025
May 2025
Year
Vehicle Price vs Units Sold
BRAND
Maruti Suzuki
Skoda
Sum of UNITSSOLD
5
4
3
1M
2M
3M
Sum of UNITPRICE_INR
Brand-wise Revenue Contribution
Increase
Decrease
Total
Total Revenue
10M
5M
0M
Skoda
Maruti Suzuki
Total
Page 1
Automobile Sales


AUTOMOBILE SALES ANALYTICS
REGION
Central
North
West
East
South
BRAND
Mahindra
Tata
Toyota
FUELTYPE
Petrol
SALEDATE
01-01-2025
25-10-2025
16.44M
Total Revenue
10.00
Total Units Sold
5
Total Sales Records
1.64M
Average Selling Price
Monthly Revenue Trend
Total Revenue
5M
0M
Mar 2025
May 2025
Jul 2025
Sep 2025
Year
Revenue by Brand and Model
Revenue by Bran...
Toyota
Tata
Mahindra
Toyota
Tata
Ma...
Fortuner
Innova ...
Nexon
Punch
Scor...
Revenue Trend Over Time
Total Revenue
6M
4M
2M
Mar 2025
May 2025
Jul 2025
Sep 2025
Year
Vehicle Price vs Units Sold
BRAND
Mahindra
Tata
Toyota
Sum of UNITSSOLD
5
0
2M
4M
6M
Sum of UNITPRICE_INR
Brand-wise Revenue Contribution
Increase
Decrease
Total
Total Revenue
20M
10M
0M
Toyota
Tata
Mahindra
Total
Page 1
Automobile Sales